

Mathematical Models in Neuroscience

LGN Neural Models

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24 May, 2025

Outline

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- Mechanistic
- Descriptive
- Interpretive
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Thalamocortical Relay Cells

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- IF Models
- HH Models

References

Types of Mathematical Models

Mathematical models in neuroscience can be categorized into three types:

- + ***Mechanistic (physics-type)***
- + ***Descriptive (statistical)***
- + ***Interpretive (normative)***

Jaeger and Jung 2015

Types of Mathematical Models

Mathematical models in neuroscience can be categorized into three types:

- ***Mechanistic (physics-type):***

Aim to find out how properties of the system arise from the physical properties of the underlying parts.

- + ***Descriptive (statistical)***

- + ***Interpretive (normative)***

Jaeger and Jung 2015

Types of Mathematical Models

Mathematical models in neuroscience can be categorized into three types:

- ***Mechanistic (physics-type):***

How neurons function?

How nervous systems operate on the basis of known anatomy, physiology, and circuitry?

- + ***Descriptive (statistical)***

- + ***Interpretive (normative)***

Dayan and Abbott 2005

Types of Mathematical Models

Mathematical models in neuroscience can be categorized into three types:

+ ***Mechanistic (physics-type)***

– ***Descriptive (statistical):***

Used to summarize experimental data or encapsulate key properties compactly.

+ ***Interpretive (normative)***

Jaeger and Jung 2015

Types of Mathematical Models

Mathematical models in neuroscience can be categorized into three types:

- + ***Mechanistic (physics-type)***

- ***Descriptive (statistical):***

 - What nervous systems do?*

 - Summarize large amounts of experimental data compactly yet accurately, thereby characterizing what neurons and neural circuits do.*

 - Their primary purpose is to describe phenomena, not to explain them.*

- + ***Interpretive (normative)***

Dayan and Abbott 2005

Types of Mathematical Models

Mathematical models in neuroscience can be categorized into three types:

- + ***Mechanistic (physics-type)***
- + ***Descriptive (statistical)***
- ***Interpretive (normative):***

Aim to model the functional roles of neural systems, for example, relating neuronal responses to the task of processing useful visual information for the animal.

Jaeger and Jung 2015

Types of Mathematical Models

Mathematical models in neuroscience can be categorized into three types:

- + ***Mechanistic (physics-type)***
- + ***Descriptive (statistical)***
- ***Interpretive (normative):***

Why neurons operate in particular ways?

Use computational and information-theoretic principles to explore the behavioral and cognitive significance of various aspects of nervous system function.

Dayan and Abbott 2005

LGN Circuit

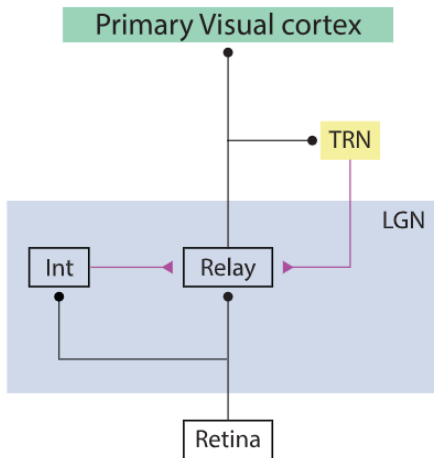
Overview of LGN Circuit

- The lateral geniculate nucleus (LGN) acts as a gateway for visual signals to reach cortex.
- It receives driving input from retinal ganglion cells (RGCs) and modulatory input from the cortex, the thalamic reticular nucleus (TRN), and the brainstem.
- The principal cells in the LGN, the thalamocortical relay cells (RCs), transmit visual information to the cortex.
- They constitute about 75–80% of the cells in the nucleus, while the remaining 20–25% are intrageniculate interneurons (INs).

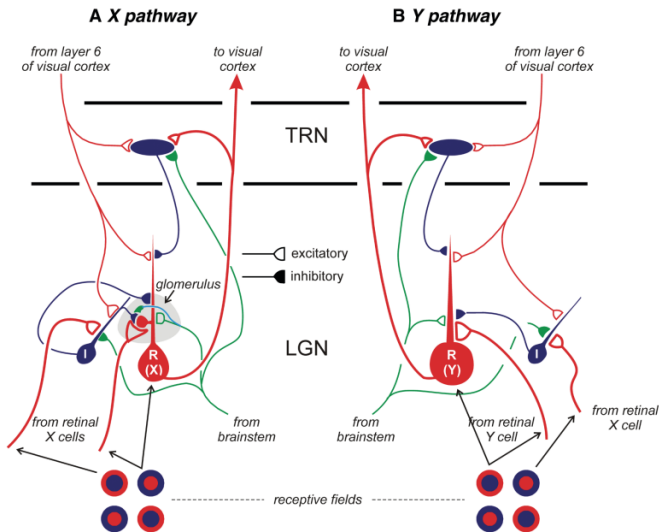
Overview of LGN Circuit

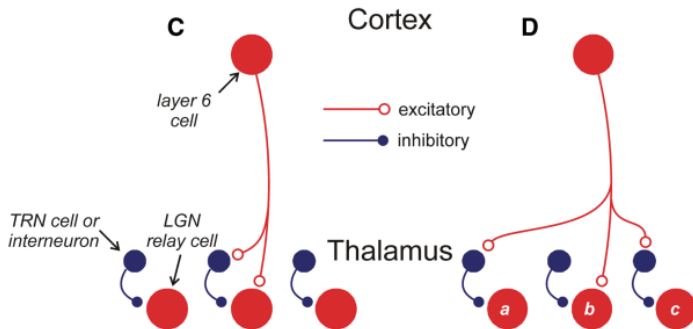
The RCs receive synaptic inputs from a variety of sources:

- Direct feedforward excitation from RGCs;
- Indirect feedforward inhibition via the INs, which in turn receive excitatory input from RGCs;
- Feedback inhibition from the TRN;
- Feedback excitation from primary visual cortex.



Simplified diagram of visual thalamic circuitry and the LGN (Adámek, Langová, and Horáček 2022).





Overview of circuitry of LGN (Shepherd and Grillner 2017)

Thalamocortical Relay Cells

Thalamocortical Relay Cells

Two prominent models of firing:

- **tonic mode**
- **burst mode**

The Importance of These Firing Patterns

Wang et al. 2025



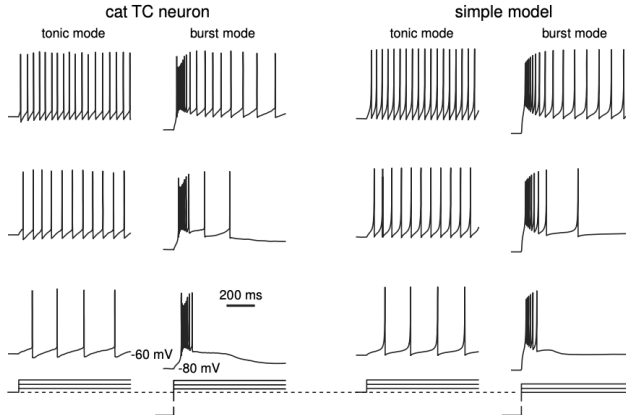
ARTICLE

Thalamic burst and tonic firing selectively indicate patients' consciousness level and recovery

Huan Wang,^{1,2,10} Yongxiang Hu,^{1,3,10} Qianqian Ge,⁴ Yuanyuan Dang,⁵ Yi Yang,⁴ Long Xu,⁴ Xiaoyu Xia,⁵ Peng Zhang,^{1,2} Sheng He,^{1,2,6} Steven Laureys,^{7,8,9} Yan Yang,^{1,2,*} and Jianghong He^{4,*}

Thalamocortical Relay Cells Firing Patterns

Izhikevich 2007



IF Models

QIF Model for Relay Cells in dLGN

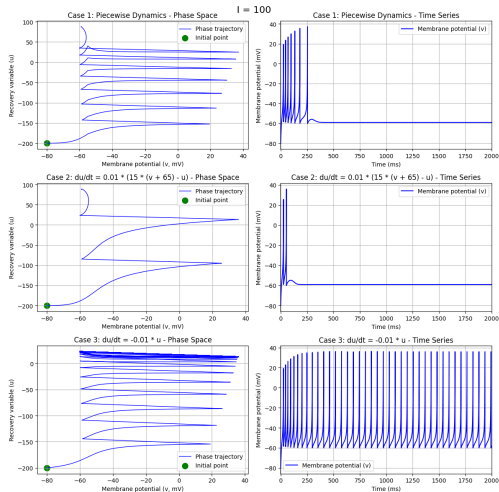
Izhikevich 2007

$$\begin{cases} 200\dot{v} = 1.6(v+60)(v+50) - u + I & v \leq -55 \\ \dot{u} = 0.01[15(v+65) - u] \\ \\ 200\dot{v} = 1.6(v+60)(v+50) - u + I & \text{else} \\ \dot{u} = -0.01u \end{cases}$$

when $v \geq 35 + 0.1u$ then $v = -60 - 0.01u$ and $u = u + 10$

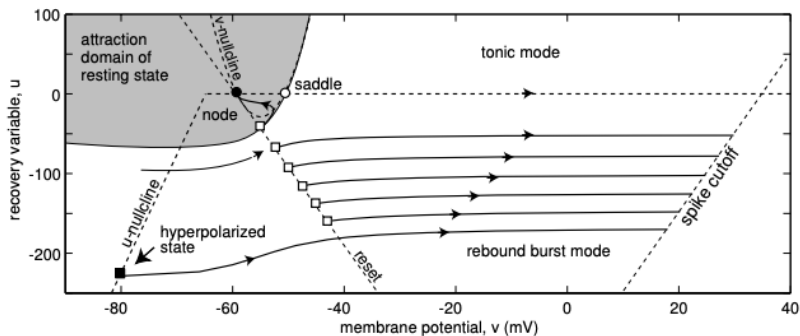
QIF Model for Relay Cells in dLGN

Izhikevich 2007



QIF Model for Relay Cells in dLGN

Izhikevich 2007



HH Models

HH-Style Model for Relay Cells in dLGN

McCormick and Huguenard 1992

JOURNAL OF NEUROPHYSIOLOGY

Vol. 68, No. 4, October 1992. Printed in U.S.A.

A Model of the Electrophysiological Properties of Thalamocortical Relay Neurons

DAVID A. MCCORMICK AND JOHN R. HUGUENARD

*Section of Neurobiology, Yale University School of Medicine, New Haven, Connecticut 06510; and
Department of Neurology, Stanford University Medical School, Stanford, California 94305*

HH-Style Model for Relay Cells in dLGN

Rhodes and Llinás 2005

A model of thalamocortical relay cells

Paul A. Rhodes, Rodolfo Llinás

First published: 22 June 2005 | <https://doi.org/10.1113/jphysiol.2004.070888> | Citations: 42

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HH-Style Model for Relay Cells in dLGN

Pospischil et al. 2008

Biol Cybern (2008) 99:427–441
DOI 10.1007/s00422-008-0263-8

PROSPECTS

Biological
Cybernetics

Minimal Hodgkin–Huxley type models for different classes of cortical and thalamic neurons

Martin Pospischil · Maria Toledo-Rodriguez ·
Cyril Monier · Zuzanna Piwkowska · Thierry Bal ·
Yves Frégnac · Henry Markram · Alain Destexhe

Received: 4 February 2008 / Accepted: 16 September 2008
© Springer-Verlag 2008

HH-Style Model for Relay Cells in dLGN

Amarillo et al. 2014

J Neurophysiol 112: 393–410, 2014.

First published April 23, 2014; doi:10.1152/jn.00647.2013.

The interplay of seven subthreshold conductances controls the resting membrane potential and the oscillatory behavior of thalamocortical neurons

Yimy Amarillo,^{1,2} Edward Zagha,² German Mato,^{1,3} Bernardo Rudy,² and Marcela S. Nadal¹

¹Consejo Nacional de Investigaciones Científicas y Técnicas, Física Estadística e Interdisciplinaria, Centro Atómico Bariloche, San Carlos de Bariloche, Rio Negro, Argentina; ²Smilow Neuroscience Program, New York University Medical Center, New York, New York; and ³Comisión Nacional de Energía Atómica, Centro Atómico Bariloche and Instituto Balseiro, San Carlos de Bariloche, Rio Negro, Argentina

Submitted 10 September 2013; accepted in final form 21 April 2014

LGN Interneurons

LGN Interneurons

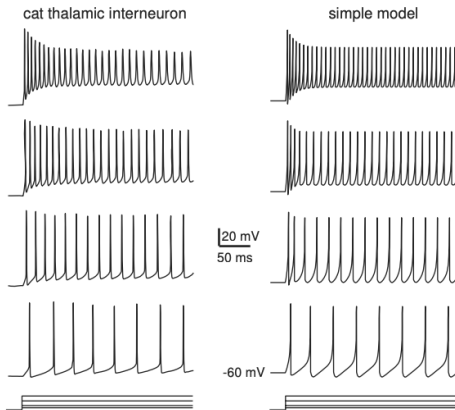
Izhikevich 2007

prominent models of firing:

- **tonic mode**

LGN Interneurons Firing Patterns

Izhikevich 2007



LGN Interneurons

IF Models

QIF Model for Interneurons in dLGN

Izhikevich 2007

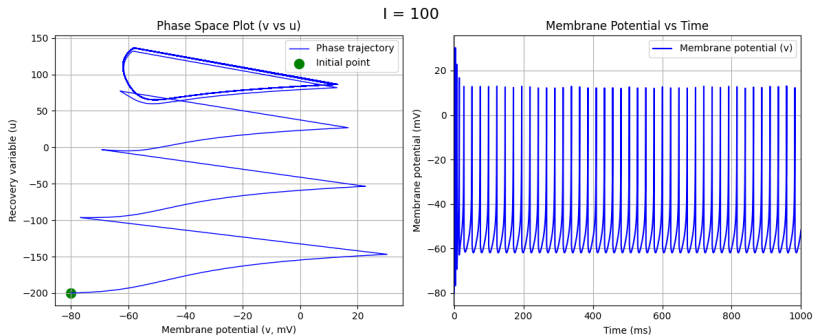
$$20\dot{v} = 0.5(v + 60)(v + 50) - u + I$$

$$\dot{u} = 0.05[7(v + 60) - u]$$

when $v \geq 20 - 0.08u$ then $v = -65 + 0.08u$ and $u = \min\{u + 50, 530\}$

QIF Model for Interneurons in dLGN

Izhikevich 2007



HH Models

A Multi-Compartment HH Model

Halnes et al. 2014

OPEN  ACCESS Freely available online

PLOS COMPUTATIONAL BIOLOGY

A Multi-Compartment Model for Interneurons in the Dorsal Lateral Geniculate Nucleus

Geir Halnes^{1*}, Sigita Augustinaite², Paul Heggelund², Gaute T. Einevoll¹, Michele Migliore³

1 IMT, Norwegian University of Life Sciences, Ås, Norway, **2** Department of Physiology, University of Oslo, Oslo, Norway, **3** Institute of Biophysics, National Research Council, Palermo, Italy

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Halnes, Geir et al. (2014). “A multi-compartment model for interneurons in the dorsal lateral geniculate nucleus”. In: *PLoS Computational Biology* 7.9, e1002160.



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Jaeger, Dieter and Ranu Jung (2015). *Encyclopedia of computational neuroscience*. Springer.

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Thank you!

